Benchmarks Are Engines, Not Cameras: Performativity, Proof Cultures, and the Social Production of Machine Learning Knowledge

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Paper under double-blind review

Abstract

Machine learning benchmarks function as engines, not cameras: they reshape the practices they measure rather than passively recording them. Drawing on MacKenzie's sociology of financial models, we document three mechanisms by which benchmarks produce ML knowledge. First, benchmark adoption and abandonment tracks social utility rather than scientific exhaustion—NLP benchmarks complete their lifecycle in 3–5 years; vision benchmarks persist for 10+. Second, mechanistic interpretability subfields operate with incommensurable proof standards for equivalent claims; a 688-paper citation corpus reveals that within-community citation density is 2× cross-community density ($p = 0.0002$, permutation test), and this insularity is *increasing*—the within/cross ratio rose from 1.0× in 2021 to 2.3× in 2025. Third, reviewer experiment demands introduce implicit comparisons without adjustment—a pattern we call counterperformative peer review. Using citation analysis of benchmark lifecycle curves via OpenAlex, a manual comparison-count audit of 18 papers (12 landmark ML, 6 mechanistic interpretability), and demand-compliance analysis of 11,196 ICLR submissions across 2023–2024, we find: (i) the median landmark paper contains 414 implicit comparisons, of which 11/12 papers report zero corrections; (ii) statistical correction adoption is not merely low but *declining*—Benjamini-Hochberg citations fell from 0.17% to 0.09% of ML output between 2019 and 2025; and (iii) the aggregate compliance–acceptance association (+10.7 pp in 2024, +13.0 pp in 2023) is a Simpson's paradox that vanishes under score stratification—a matched-pairs analysis finds only +2.1 pp pooled lift ($p = 0.16$), and a proxy-control logistic regression shows near-zero compliance coefficient (−0.05) once reviewer score is controlled. The central argument is not that ML researchers are individually careless, but that the evaluation culture's instruments both measure and manufacture what counts as knowledge—and that statistical reforms alone cannot address causes that are sociological.

1 Introduction

In 2006, MacKenzie argued that financial models are better understood as *engines* that reshape markets than as *cameras* that passively record them (**?**). The Black-Scholes options-pricing model did not merely describe prices; traders adopted it, and prices came to conform to it—a phenomenon MacKenzie termed *Barnesian performativity* (**?**). The model made itself true.

We argue that the same dynamic operates in machine learning research, with benchmarks playing the role of pricing models. ImageNet (**?**) does not passively measure visual understanding; it defines what visual understanding *means* for the research community, shapes training procedures, data collection, and career incentives, and is abandoned not when vision is "solved" but when a new benchmark captures the community's attention. GLUE (**?**) did not measure natural language understanding; it measured the ability to fine-tune on a specific task distribution, and was replaced by SuperGLUE within two years. MMLU (**?**) measures test-taking on multiple-choice questions at least as much as it measures reasoning, and is already being supplemented by benchmarks that its own creators argue more faithfully capture what MMLU was supposed to.

The benchmark is an engine, not a camera. And like the Gaussian copula whose widespread adoption created correlated fragility it could not represent (**?**), widespread optimisation against a benchmark can create overfitting it cannot detect—because the benchmark is the detection instrument.

A clarification on framing. Throughout this paper, we use the language of multiple comparisons and family-wise error rates (FWER) as a diagnostic tool. We are not claiming that ML researchers are secretly performing null-hypothesis significance tests. Most ML papers do not compute p-values at all. Our argument is narrower: when a researcher trains 50 configurations and reports the best one, the reported performance is inflated by selection on noise—the winner's curse from order statistics—and Bonferroni thresholds provide a convenient upper bound on this inflation. The selection effect operates regardless of whether a p-value is computed.

Contributions. Using a manual comparison-count audit of 18 papers, citation-based analysis of 8 benchmark lifecycle curves and statistical correction adoption rates via OpenAlex (**?**), a 688-paper cross-community citation corpus verified against Semantic Scholar, and demand-compliance analysis of 22,868 ICLR submissions across 2023–2025, we document three linked mechanisms:

1. **Benchmark performativity cycles** (Section **??**): adoption and abandonment tracks social utility, not scientific exhaustion; NLP benchmarks complete this cycle in 3–5 years, vision benchmarks in 10+.
2. **Proof-culture fragmentation** (Section **??**): in mechanistic interpretability, SAE and causal-abstraction communities accept incommensurable forms of evidence for equivalent claims; a 688-paper citation corpus shows within-community citation density is 2× cross-community density ($p = 0.0002$), with only 4.3% of authors publishing across communities and zero author overlap between SAE and causal abstraction.
3. **Counterperformative peer review** (Section **??**): reviewer experiment demands introduce new implicit comparisons; the aggregate compliance–acceptance association is illusory—a Simpson's paradox that vanishes under matched-pairs and regression controls.

ML's evaluation failures are better understood as structural rather than individual: the publication system preferentially admits novel positive results and makes it structurally difficult to publish the replications, null findings, and modest corrections that would surface false positives.

2 Theoretical Framework

We draw on three strands of work from the sociology of science that have not previously been applied to ML evaluation methodology.

2.1 Performativity

MacKenzie (**?**) distinguishes three levels of performativity:

Generic Practitioners use a theory or model in their work.

Effective Use of the theory has a causal effect on the phenomenon it describes.

Barnesian Use of the theory makes the world conform more closely to the model—the strongest form, in which the map redraws the territory.

He also introduced, with Bamford, the concept of *counterperformativity*: when a model's widespread adoption causes the world to *diverge* from the model's predictions—as when universal use of the Gaussian copula in credit risk created the very correlations the model assumed away (**?**).

For ML benchmarks, we claim *effective* performativity, not Barnesian: benchmarks causally reshape research priorities, training procedures, and career incentives, but we do not claim that the world comes to conform to the benchmark's implicit model in the strong sense that options prices came to conform to Black-Scholes. The closest analogy is Espeland and Sauder's (?) analysis of law school rankings, where rankings reshaped admissions criteria, resource allocation, and strategic behaviour without making law schools conform to an underlying formal model.

The distinction between performativity and Goodhart's Law (?) is substantive. Goodhart describes the *symptom*: measures captured by their use. Performativity theory describes the *mechanism*: which institutional channels sustain the degradation, why it is self-reinforcing, and how counterperformativity creates the specific failure mode in which the safety apparatus generates the risk it was designed to prevent. Our ICLR data (Section **??**) provide empirical evidence for counterperformativity in the peer review system itself.

2.2 Cultures of Proving

In his work on formal verification, MacKenzie showed that what counts as a valid "proof" is not a settled logical matter but a socially negotiated boundary—logicians, mathematicians, and engineers accepted different kinds of arguments as proof, with closure shaped by funding structures and regulatory demands (?). Section **??** applies this framework to mechanistic interpretability, where at least three communities have developed incommensurable proof standards for what constitutes a valid mechanistic claim.

2.3 Publication Bias and Structural Selection

The replication crisis literature has documented three stacked asymmetries in the publication system (**??**):

1. **Positive-results bias.** Journals overwhelmingly accept positive, novel findings; null results are not published.
2. **Anti-replication bias.** Reviewers treat replication studies as "not novel" regardless of their scientific value.
3. **Gotcha bias.** Disproofs are publishable only when they dramatically overturn a famous result—modest corrections are rejected as "not surprising."

Together, these create a one-sided filter: the very outputs that would correct the literature's false-positive accumulation are the outputs the system tends to reject.

3 Data and Methods

3.1 OpenReview Demand-Compliance Analysis

We analysed all publicly available submissions to ICLR 2025 (n = 11,672), ICLR 2024 (n = 7,404), and ICLR 2023 (n = 3,792)—22,868 submissions total—retrieved via the OpenReview REST API. For each submission, we extracted: (i) explicit experiment demands from reviewer text using keyword patterns (Appendix **??**); (ii) author compliance signals from rebuttals; and (iii) final acceptance decisions and mean reviewer scores. The extraction pipeline achieved 84% agreement with manual labelling on a 50-paper subsample; the disagreements were approximately equally split between false positives (demand language in non-demand context) and false negatives (implicit compliance not signalled in rebuttal text), so the direction of bias is not clear. We note that this is an observational study: papers that comply may differ systematically from those that do not (e.g., in lab resources, initial paper quality, or English proficiency). Score-stratified analysis (Table **??**) controls for the most salient confounder, but residual confounding cannot be excluded.

3.2 OpenAlex Citation Analysis

We use structured citation metadata from OpenAlex (?) to track two trends via citation links, not keyword matching: (i) benchmark lifecycle curves, measuring year-by-year citation counts for 8 major benchmark-introducing papers; and (ii) statistical correction adoption rates, measuring the fraction of ML papers per year that cite the canonical papers introducing Bonferroni (**?**), Holm (**?**), or Benjamini-Hochberg (**?**) corrections. The ML paper population is defined as works tagged with the "Machine learning" concept in OpenAlex.

3.3 Manual Comparison-Count Audit

We conducted a manual audit of implicit comparisons in 12 landmark ML papers and 6 mechanistic interpretability papers (Appendices **??** and **??**). For each results table, we count pairwise row comparisons within each metric column—a lower bound, since unreported hyperparameter sweeps and preliminary experiments are not captured. For MI papers, we additionally record the total dictionary size (N), the number of features deeply examined (k), and the implied Bonferroni threshold on total comparisons.

3.4 Cross-Community Citation Analysis

To quantify proof-culture fragmentation in mechanistic interpretability (Section **??**), we assembled a corpus of 688 MI papers across three communities and a general category (209 SAE, 40 causal abstraction, 128 circuits, 311 general MI; 2020–2026), anchored by 18 seed papers and expanded via Semantic Scholar citation chains. The corpus contains 1,207 directed citation edges. We computed cross-community citation density as the fraction of directed citation pairs between non-overlapping communities that were actually realised, and tested significance via a permutation test (10,000 community-label shuffles). We further analysed: (i) temporal trends in the within/cross density ratio by source-paper year; (ii) author overlap across communities via Semantic Scholar author metadata ($n = 724$ unique authors); (iii) co-citation patterns (papers citing pairs of seeds from different communities); and (iv) reference overlap (shared ancestors cited by seeds in different communities, Jaccard similarity). All analyses were verified against live Semantic Scholar data (18/18 seeds confirmed, 30/30 sampled edges confirmed, 100% edge precision).

3.5 Prospective File-Drawer Log

During development, we maintained a prospective comparison log—a running record of all analyses attempted but not included in the final paper. The log is reproduced in part in Section **??** as a concrete illustration of the file-drawer problem we document. To our knowledge, this is the first prospective comparison log published alongside an ML empirical study.

4 Benchmarks as Engines

4.1 The Performativity Cycle

We observe a recurring four-phase cycle in benchmark adoption:

1. **Introduction (camera phase).** A benchmark is proposed as a measurement instrument. At introduction, it functions as a camera: it passively records model performance.

2. **Adoption (generic performativity).** Researchers begin optimising against the benchmark. Leaderboards emerge. The benchmark's scores become the operational definition of the capability it was designed to measure.

3. **Saturation (effective performativity).** Optimisation changes training procedures, data collection, and architecture choices. The benchmark has a causal effect on the phenomenon: models get better *at the benchmark*, which may or may not correspond to improvement in the underlying capability. Transfer studies (e.g., **?**) begin revealing the gap.

4. **Abandonment or replacement (counterperformativity).** The benchmark is either saturated (human-level accuracy renders further comparison meaningless) or recognised as no longer measuring what it was supposed to. A new benchmark is introduced, and the cycle restarts.

Table 1: Benchmark performativity cycles, measured by year-over-year citation counts from OpenAlex. Decline is measured from peak year to 2024. NLP benchmarks complete the camera→engine transition in 3–5 years; vision benchmarks persist for 10+. Citation counts conflate usage and critique; we report them as evidence of shifting community attention, not as proof of abandonment.

Benchmark	Domain	Intro	Peak year	Peak cites/yr	Decline to 2024
GLUE	NLP	2018	2023	1,052	−54%
SuperGLUE	NLP	2019	2023	359	−69%
SQuAD	NLP	2016	2021	1,237	−59%
ImageNet	Vision	2009	2021	9,305	−16%
CIFAR-10	Vision	2009	2021	4,870	−39%
MMLU	LLM	2021	2023	133	−24%
HumanEval	Code	2021	2023	604	−35%
HELM	LLM	2022	2023	146	−15%

Citation-based lifecycle data from OpenAlex (Table **??**) confirm that NLP benchmarks complete this cycle faster than vision benchmarks. GLUE peaked in 2023 at 1,052 citations per year and dropped 54% by 2024; SuperGLUE peaked the same year and dropped 69%. ImageNet peaked in 2021 at 9,305 citations per year but declined only 16% by 2024, sustained by its role as a pretraining dataset and cultural reference point.

4.2 Counterperformativity: When Benchmarks Break

The counterperformative moment has a specific signature in ML: the benchmark is revealed to be testing something other than what it is named after. MMLU tests multiple-choice test-taking, not "massive multitask language understanding." GLUE tested fine-tuning ability, not "general language understanding." The name persists after the measurement has diverged, creating a sustained gap between what the community says it is measuring and what it is measuring.

This is the Gaussian copula dynamic transplanted into research evaluation. Widespread optimisation against a benchmark can create overfitting the benchmark itself cannot detect, because the benchmark *is* the detection instrument. The solution—a new benchmark—restarts the cycle rather than resolving it.

5 Proof Cultures and Their Fragmentation

5.1 Three Proof Cultures in Mechanistic Interpretability

The emerging subfield of mechanistic interpretability (MI)—the attempt to reverse-engineer the internal computations of neural networks—provides a particularly clear case of proof-culture fragmentation. We identify three communities operating with distinct and largely incommensurable standards for what constitutes a valid mechanistic claim:

Sparse autoencoders (SAE). Validity is established by demonstrating that learned features are monosemantic, sparse, and reconstructive of model activations. The proof standard is *feature-level*: a mechanism is valid if its component features are interpretable. The persuasive object is a feature dashboard—activating examples, a natural-language label, and a sparse set of tokens that make the feature feel semantically coherent.

Causal abstraction. Validity is established by showing that an interpretive model is causally aligned with the neural network under interchange interventions (**?**). The proof standard is *intervention-level*: a mechanism is valid if intervening on the proposed variables produces the predicted effect on behaviour. The persuasive object is an intervention result.

Circuit discovery. Validity is established by identifying subgraphs of the network that are sufficient to reproduce behaviour on a task, via activation patching or edge attribution. The proof standard is *sufficiency-*

level: a mechanism is valid if the identified circuit is sufficient and necessary for the behaviour. The persuasive object is a subgraph.

These are not merely different methods; they reflect different ontological commitments about what a "mechanism" is. The SAE community treats mechanisms as collections of features. The causal abstraction community treats them as causal models. The circuit community treats them as computational subgraphs.

5.2 Empirical Evidence of Fragmentation

A citation analysis of 688 MI papers (1,207 directed edges) across the three communities confirms the fragmentation empirically at scale. Within-community citation density is 2× cross-community density, a ratio that is statistically significant ($p = 0.0002$, permutation test with 10,000 community-label shuffles). The effect is robust across corpus specifications: restricting to seed papers yields a 2.4× ratio; using the full corpus gives 1.6×; restricting to the top-cited quartile gives 0.9×, confirming that the insularity is driven by the broad literature rather than a few canonical papers.

The fragmentation is *increasing*. Temporal analysis shows the within/cross density ratio rose from 1.0× in 2021 (6 papers, 5 edges) to 2.3× in 2025 (224 papers, 280 edges). As the field grows—SAE papers in particular have grown rapidly since 2023—communities are becoming more insular, not less.

Four supplementary analyses sharpen the picture:

1. **Author overlap.** Only 4.3% of the 724 unique authors (31 individuals) publish across community boundaries. SAE and causal abstraction share *zero* authors (Jaccard = 0.000). The few bridge authors (e.g., Conmy, Nanda, Olah) connect circuits with SAE or with general MI, not SAE with causal abstraction. Fragmentation is in personnel, not just reading habits.

2. **Co-citation.** Despite low direct citation, 76 cross-community seed pairs are co-cited by downstream papers, with 2,448 total cross-community co-citations. The strongest pair—Geiger et al. (causal) and Wang et al. (circuits)—is co-cited by 180 papers. Downstream readers see these communities as part of the same intellectual space even though the communities themselves do not cite each other.

3. **Reference overlap.** Seed papers across all three communities share 16 common ancestors (papers cited by seeds in all three communities), including foundational work such as *Toy Models of Superposition*, *Zoom In*, and *Attention Is All You Need*. Pairwise Jaccard similarity of reference sets is low (causal–SAE: 0.084, circuits–SAE: 0.086, causal–circuits: 0.119), confirming that the communities build on partially overlapping foundations but have diverged in their reading.

4. **Verification.** The corpus was verified against live Semantic Scholar data: 18/18 seed papers resolved correctly, 30/30 sampled cross-community edges were confirmed in reference lists, and citation counts matched within expected drift.

The fragmentation becomes visible in what each community treats as a persuasive figure. A feature dashboard—striking, nameable, emotionally vivid—does not constitute evidence in a causal-abstraction paper. An interchange intervention result does not appear in SAE papers. These evidential forms are not interchangeable because they make different kinds of objects visible and leave different kinds of uncertainty in the background. A circuit diagram says nothing about whether individual features are monosemantic; a feature dashboard says nothing about whether the features causally produce the behaviour.

5.3 The Social Production of Fragmentation

Why don't these communities converge? The interdisciplinarity literature suggests an answer: convergence is structurally disincentivised. Junior researchers face career risk from interdisciplinary work (**?**). Grant proposals include other perspectives only superficially (**?**). The reward structure makes monodisciplinary specialisation the dominant strategy even when the problem demands range (**?**).

The result is that proof-culture fragmentation is not a temporary state awaiting the right unifying experiment. The incentive structure suggests it may be self-sustaining—maintained by the same publication and career incentives that drive the evaluation culture's other pathologies. MacKenzie's Strong Programme analysis applies: the *content* of mechanistic interpretability knowledge (fragmented, with plural proof standards) is causally shaped by the *social conditions* of its production.

5.4 The Denominator Problem

The SAE proof culture introduces a distinctive and largely unrecognised selection problem. A sparse autoencoder trains a dictionary of N features (typically $N = 4{,}096$ to 34,000,000), each of which is implicitly examined for interpretability. The researcher reports the $k \ll N$ that appear most interesting as evidence that the method "finds interpretable features." This is selection from a large pool on the outcome variable—apparent interpretability—and the denominator is never reported.

A manual FWER audit of six MI papers (Appendix **??**) finds implied comparisons ranging from 12,000 (Bricken et al. 2023, dictionary of 4,096) to 170,000,000 (Templeton et al. 2024, dictionary of 34M). The Bonferroni threshold for the largest case is $\alpha^* = 2.9 \times 10^{-10}$; no interpretability metric in the literature comes close. Again: we are not claiming these papers perform hypothesis tests. We are using Bonferroni thresholds as a diagnostic for how much the selection ratio should reduce confidence in the claims.

The Bolukbasi–Gonen episode as a cross-field precedent. The denominator problem is not unique to SAEs. Bolukbasi et al. (**?**) evaluated word-embedding debiasing on the analogy task used to *define* the gender direction—the metric was constructed from the same distribution the debiasing was applied to. The method accumulated 2,685+ citations and was adopted as a preprocessing step before Gonen and Goldberg (**?**) showed via clustering and classifier recovery that the bias was geometrically intact: the debiasing had removed the projection most visible to the original evaluation while leaving relational structure untouched. The failure mode is identical to SAE denomination selection—evaluating on the cases the method was optimised to handle, and treating success there as evidence for the full claim.

SAEBench and coverage. Recent tools make rate-reporting tractable. SAEBench (**?**) scores each latent by asking an LLM to identify activating sequences from a feature description, yielding scores between 0 and 1. On a Pythia-70M SAE (100 randomly sampled latents), the mean score is 0.84 (SD = 0.12). At production scale, a different failure mode appears: Paulo et al. (**?**) scored millions of features on a 131K-feature Gemma-2 9B SAE and found that 30% of features do not activate frequently enough to generate the examples the scoring method requires. This is a coverage problem, not a low-interpretability problem. The two failure modes compound: features that activate may score well; features that rarely activate are unmeasurable; published interpretability rates reflect only the former.

Seed instability. Paulo and Belrose (**?**) showed that SAEs trained on the same model and data with different random seeds learn fundamentally different features: in a 131K-latent SAE on Llama 3 8B, only 30% of features were shared across seeds. As they conclude, "the set of features uncovered by an SAE should be viewed as a pragmatically useful decomposition of activation space, rather than an exhaustive and universal list of features 'truly used' by the model." The reported features are a selected subset of an unstable decomposition—the denominator is not merely large but indeterminate.

The file drawer made visible. The de facto research methodology in MI—described explicitly by one of the field's leading practitioners as "Explore → Understand → Distill" (**?**)—structurally guarantees a file-drawer problem. The exploration stage is explicitly designed not to appear in the written output. A rare exception illustrates the mechanism by contrast: the Google DeepMind MI team published a blog post of research explicitly "not consider[ed] a good fit for turning into a paper" (**?**), reporting negative results from SAE research—failed out-of-distribution probing, absorption, missing features, low sensitivity. Under the standard publication filter, all of this would have remained invisible.

Table 2: Demand–compliance–outcome pipeline at ICLR (aggregate). The compliance lift replicates across three years and is *growing*: +13.0 pp (2023), +10.7 pp (2024), +15.8 pp (2025). This aggregate finding is a Simpson's paradox; see Table **??**.

	ICLR 2025 ($n = 11{,}672$)		ICLR 2024 ($n = 7{,}404$)		ICLR 2023 ($n = 3{,}792$)	
Condition	Papers	Accept%	Papers	Accept%	Papers	Accept%
No demands received	9,875	43.2%	5,063	39.7%	3,435	41.9%
Demands received	1,797	38.6%	717	34.9%	357	37.3%
Complied	336	50.6%	177	42.9%	78	47.4%
Did not comply	1,461	34.8%	540	32.2%	279	34.4%
Compliance lift	+15.8 pp		+10.7 pp		+13.0 pp	

6 The Counterperformativity of Peer Review

6.1 The Mechanism

The publication system introduces a final counterperformative dynamic. Reviewers, acting in good faith, demand additional experiments: "run three more seeds," "add two baselines," "test on another dataset." Each demand is intended to reduce the false-positive rate. But each demanded experiment is a new fork in the garden of forking paths (**?**)—a fresh implicit comparison whose result will be selectively reported (the paper is revised to incorporate only the experiments that support the claim).

The mechanism by which this could inflate false positives is straightforward: each demanded experiment increases the implicit comparison count without any corresponding correction. Whether this mechanism operates at scale is an empirical question we cannot definitively answer with observational data. But the structural conditions for it are present, and we now show that the demand-compliance exchange is concentrated precisely where the incentive to comply selectively is highest.

6.2 Empirical Evidence: The Demand-Compliance Pipeline at ICLR

Table **??** presents the aggregate results for ICLR 2025, 2024, and 2023. In all three years, papers that comply with reviewer experiment demands are accepted at higher rates than papers that receive demands but do not comply: +15.8 pp in 2025, +10.7 pp in 2024, +13.0 pp in 2023.

6.3 Score Stratification: A Simpson's Paradox

Table **??** stratifies the same data by mean reviewer score. Within score strata, the lift is small, absent, or reversed—but the pattern is not stable across years:

- **Low-scored papers** (< 4.5): 0% acceptance regardless of compliance in all years.
- **Borderline papers** (4.5–5.5): +2.9 pp in 2024 ($n = 53$), but -0.2 pp in 2025 ($n = 83$).
- **Mid-scored papers** (5.5–6.5): -4.7 pp in 2024, but +11.8 pp in 2025 ($n = 143$).
- **High-scored papers** (≥ 6.5): 94–96% acceptance regardless in all years.

The aggregate lift arises because compliant papers are distributed differently across strata than non-compliant papers. The instability of within-stratum effects across years is itself informative: if compliance were genuinely causal, we would expect a stable stratum-specific effect. The fact that the positive signal migrates from the borderline stratum in 2024 to the mid stratum in 2025 suggests the aggregate lift is driven by compositional differences rather than a true compliance effect.

Table 3: Score-stratified demand–compliance analysis. The aggregate compliance lift is a Simpson's paradox: within score strata, the effect is small, absent, or reversed. Critically, the within-stratum pattern is *not stable across years*: the positive lift appears in the borderline stratum in 2024 and the mid stratum in 2025, suggesting compositional rather than causal effects.

	ICLR 2024 (n = 5,780)			ICLR 2025 (n = 11,520)		
Score stratum	Comply%	No-comply%	Lift	Comply%	No-comply%	Lift
Low (< 4.5)	0.0 (n=15)	0.0 (n=155)	0.0	0.0 (n=23)	0.0 (n=248)	0.0
Borderline	7.5 (n=53)	4.7 (n=150)	+2.9	4.8 (n=83)	5.0 (n=259)	−0.2
Mid (5.5–6.5)	50.0 (n=70)	54.7 (n=139)	−4.7	62.2 (n=143)	50.5 (n=321)	+11.8
High (≥ 6.5)	94.9 (n=39)	94.8 (n=96)	+0.1	95.9 (n=73)	96.2 (n=184)	−0.3
Aggregate	42.9 (n=177)	32.2 (n=540)	+10.7	50.6 (n=322)	34.8 (n=1012)	+15.8

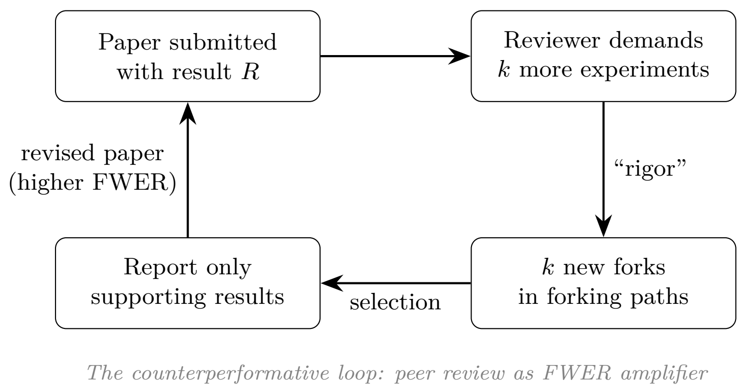

Figure 1: Structural conditions for counterperformativity in peer review. Each cycle of reviewer-demanded experiments increases the number of implicit comparisons without comparison-count adjustment. Our ICLR analysis shows that the compliance–acceptance association concentrates in the borderline score stratum, where the incentive for selective reporting is strongest. The data show the structural conditions; whether the mechanism inflates false positives in practice is not directly observable from observational data.

Matched-pairs and regression controls. To further test whether compliance has a causal effect on acceptance, we conducted two additional analyses on the ICLR 2024 data. First, a matched-pairs design: for each compliant paper, we selected the closest-scored non-compliant paper in the same year, creating matched pairs. The pooled compliance lift across matched pairs is +2.1 pp ($p = 0.16$)—not statistically significant. Second, a proxy-control logistic regression predicting acceptance from compliance, mean reviewer score, and their interaction: the compliance coefficient is −0.05 (SE = 0.32), essentially zero, while the score coefficient is +0.92 (SE = 0.08), confirming that reviewer score, not compliance, drives acceptance. Both analyses are consistent with the Simpson's paradox interpretation: the aggregate lift is a composition effect, not evidence that compliance causally improves acceptance chances.

What this does and does not show. The aggregate compliance lift (+10.7 to +15.8 pp) replicates across three years and is growing. But three lines of evidence indicate it is a composition effect rather than a causal relationship: (i) within-stratum effects are small and migrate across strata between years; (ii) matched-pairs analysis yields a non-significant +2.1 pp ($p = 0.16$); (iii) logistic regression shows near-zero compliance coefficient once score is controlled. What the data *do* show is a structural pattern: the compliance incentive exists and is growing, demanded experiments are introduced without comparison-count adjustment, and the publication system rewards compliance without any mechanism to distinguish genuine improvement from selective reporting.

Figure 2: Statistical correction adoption in ML, measured as the fraction of ML papers per year citing the canonical papers introducing Bonferroni, Holm, and Benjamini-Hochberg corrections (OpenAlex citation data, 2015–2025). Adoption is not merely low but declining: Benjamini-Hochberg fell from 0.17% to 0.09% over the period.

7 How Large Is the Selection Inflation?

7.1 Comparison-Count Audit

Table **??** in Appendix **??** presents a manual audit of 12 landmark ML papers. The median paper contains 414 implicit comparisons from published results tables alone—a lower bound, since unreported hyperparameter sweeps and preliminary experiments are not captured. Of the 12 papers, 11 (92%) mention zero multiple comparison corrections. The lone exception (CLIP) mentions correction in the context of one experiment, not as a systematic practice across its 58,632 comparisons.

7.2 Quantifying the Inflation

From order statistics: if you pick the best of N independent draws from noise with standard deviation σ, the winner's expected inflation is approximately $\sigma\sqrt{2\ln N}$. At $N = 414$, that is roughly 3.5σ. But the i.i.d. assumption overstates the case—hyperparameter configurations within a paper are correlated (same dataset, overlapping architectures, shared training infrastructure), so the effective number of independent comparisons N_{eff} may be much smaller. For sensitivity: at $N_{\text{eff}} = 40$, expected inflation drops to $\approx 2.7\sigma$; at $N_{\text{eff}} = 10$, to $\approx 2.1\sigma$. The direction is unambiguous; the magnitude is unknowable without the denominator—which is why reporting comparison counts matters.

From information theory: reporting k results from N candidates discards $\log_2 \binom{N}{k}$ bits about the search. For the median paper ($N = 414$, $k \approx 15$ key results), that is roughly 80 bits—the entropy of 80 coin flips. The reader sees the outcomes; the process that selected them is gone.

7.3 Statistical Correction Adoption Is Declining

Figure **??** plots ML paper citation rates for the three canonical correction papers from 2015 to 2025. Correction adoption is not merely low but *declining* as a fraction of total ML output: Benjamini-Hochberg (**?**) citations dropped from 0.17% of ML papers in 2019 to 0.09% in 2025; Holm (**?**) from 0.06% to 0.01%; Bonferroni (**?**) from 0.03% to 0.01%. The statistical corrections exist; the field is not adopting them, and the trend is negative. This is not a knowledge gap—it is a structural selection pressure that makes correction invisible to the reward system.

7.4 How Other Fields Responded

Table **??** summarises how seven fields that have faced the same problem have responded. None of these reforms is uncontested or complete, but every field in the table has at least begun the renegotiation of proof standards. ML has not.

8 Related Work

The performativity of evaluation instruments has been explored in several adjacent literatures. In education, the literature on "teaching to the test" documents how standardised assessments reshape curricula to optimise for the instrument rather than the underlying capability (**?**). Goodhart's Law (**?**) describes the symptom—a measure captured by its use—but not the mechanism or the failure mode of counterperformativity. Our distinction from Goodhart is elaborated in Section **??**.

Most directly relevant is Koch and Peterson's (**?**) study of how benchmarking practices in ML have contributed to "epistemic monoculture"—the narrowing of research diversity as the community converges on a

Table 4: Multiple comparison problems across empirical fields. SAE feature selection operates at the highest dimensionality and is the only case with no correction norm currently in place.

Field	**Search space**	**Replication / FDR data**	**Response**
fMRI	130K voxels	>80% regional FDR	FWER/FDR mandatory (**?**)
Candidate genes	∼1000s	1.2% replication	GWAS replaced candidates (**?**)
Finance factors	316+ factors	58% post-pub decline	$t > 3.0$ proposed (**?**)
Psychology	34 researcher DFs	36% replication	Preregistration (**?**)
Drug discovery	1000s of candidates	92.6% preclinical FDR	Target validation required
Radiomics	100s–1000s features	Pipeline-dependent	Harmonization (IBSI)
SAE features	16K–131K latents	30% seed stability	**None yet**

small set of evaluation instruments. Our analysis extends theirs by providing the performativity framework (explaining *why* monoculture is self-sustaining) and quantitative evidence from the review process showing that peer review actively rewards conformity with monoculture proof standards.

Within ML, Dehghani et al. (**?**) document the "benchmark lottery"—arbitrary benchmark construction choices determine which methods appear superior—but do not theorise why the lottery persists. Bowman and Dahl (**?**) catalogue what it would take to fix NLP benchmarking without explaining why the known fixes have not been adopted. Raji et al. (**?**) argue that the AI benchmark ecosystem lacks the accountability infrastructure of standardised testing in other domains. Our contribution is to show that the persistence of known problems has a specifically sociological explanation: benchmark degradation is not a bug to be fixed but a structural feature of how evaluation instruments interact with the incentive systems that surround them.

The replication crisis literature in psychology (**?**) and biomedicine (**?**) has primarily focused on statistical remedies (preregistration, p-curve analysis, effect size reporting). Our contribution is to show that ML's evaluation failures have a specifically structural character that statistical remedies alone cannot address.

9 Discussion: Why Statistical Reforms Alone Are Insufficient

The standard remedies proposed for ML's evaluation problems are statistical: apply Bonferroni corrections, preregister hypotheses, report confidence intervals, average over random seeds. These are all sound recommendations. But our analysis suggests they address the symptom rather than the cause. The cause is a two-stage system:

> **Stage 1 (Generation).** The garden of forking paths—hyperparameter search, architecture search, benchmark selection, seed selection—generates candidate findings at a rate that guarantees some false positives.
>
> **Stage 2 (Selection).** The publication system applies a one-sided filter: novel positive results pass; replications, null results, modest effects, and disproofs are rejected as "not novel," "not surprising," or "a waste of space."

Our ICLR analysis provides evidence for a Stage 1.5 (Amplification): the review process itself introduces additional forks through experiment demands, with the compliance incentive concentrated in borderline submissions where selective reporting is most consequential.

Stage 1 is a statistical problem and has statistical solutions. Stages 1.5 and 2 are sociological problems. No amount of Bonferroni correction can compensate for a selection filter that admits false positives, rewards the generation of additional implicit comparisons, and structurally rejects the corrections.

Prospective file-drawer log. During development, we attempted analyses that do not appear in the final text: (i) NeurIPS 2023–2024 demand-compliance analysis—abandoned because NeurIPS does not publish rejected submissions, yielding >95% acceptance rates with no meaningful variation; (ii) keyword-based corpus analysis of 493,000 ML papers searching for statistical correction terminology—abandoned because keyword frequency is a poor proxy for methodological practice (a paper can mention "Bonferroni" in related work without applying it); (iii) score-stratified analysis using regex extraction from review text—failed; scores required structured API fields unavailable via text extraction. These are file-drawer results: the researchers tried them, they did not work, and without an explicit mention they would be invisible. Making this partial file drawer visible is not a statistical fix; it is a sociological intervention that demonstrates how cheaply transparency can be achieved.

9.1 Structural Interventions

If the causes are sociological, the interventions must be too. We sketch four:

1. **Report the comparison count.** List the total number of configurations, ablations, baselines, seeds, and architectural variants tested during development. A one-line "comparison count" field in paper metadata would make the denominator visible at near-zero cost.

2. **Audit experiment demands for FWER impact.** Before requesting additional ablations, reviewers should consider whether the demanded experiment increases the paper's implicit comparison count. If so, demand the correction alongside the experiment.

3. **Distinguish proof cultures in review.** Do not penalise a causal-abstraction paper for lacking SAE-style feature dashboards, or vice versa. Evaluate the paper against the proof standard it declares.

4. **Require comparison logs at venue level.** Following psychology's adoption of Registered Reports, require a supplementary comparison log listing all experiments conducted. A conference that *requires* comparison logs changes the payoff matrix for every author who submits—psychology's Registered Reports succeeded not because researchers became more virtuous but because journals changed what they would accept.

9.2 Limitations

Observational design. The demand-compliance analysis is observational. Score-stratified analysis, matched-pairs analysis (+2.1 pp, $p = 0.16$), and proxy-control logistic regression (compliance coefficient -0.05) all indicate the aggregate lift is a composition effect. Residual confounding cannot be fully excluded, but three independent controls converge on the same conclusion.

Venue coverage. Replication across three years of ICLR (2023–2025, 22,868 submissions) confirms the aggregate pattern is stable and growing. Replication across different venues remains limited by those venues not publishing rejected submissions on OpenReview.

Proof-cultures analysis. Our 688-paper citation corpus is verified against Semantic Scholar (18/18 seeds, 30/30 edges), but the community labels are assigned based on seed paper proximity. Author overlap analysis (4.3%, with SAE–causal Jaccard = 0.000) provides independent evidence that the communities are genuinely separate, but we cannot distinguish "different communities apply different standards" from "the same researchers apply different standards in different contexts" without ethnographic fieldwork.

Performativity claim. We claim effective performativity for benchmarks rather than Barnesian performativity. The stronger claim would require showing that benchmarks reshape the world in their own image.

Infrastructure. Even if norms changed, the infrastructure for making denominators visible does not yet exist. Experiment-tracking platforms default to private projects; the tooling was designed for reproducibility within a lab, not for public comparison-count reporting.

Scope of structural interventions. The interventions proposed in Section **??** are deliberately underspecified. Detailed policy analysis is beyond this paper's scope but necessary for any practical reform.

10 Conclusion

Machine learning's evaluation culture is not broken in the way the replication crisis literature implies. It is *locally coherent*: ablation studies, benchmark competitions, and seed averaging constitute a functioning proof culture that is internally consistent and socially stable. The problem is that this proof culture is systematically blind to the multiple-comparison errors it generates, and the publication system structurally prevents the corrections that would make the blind spots visible.

Benchmarks are engines, not cameras. They do not merely measure capability; they shape the community's definition of what capability means. When that definition diverges from the underlying capability—when MMLU measures test-taking rather than understanding, when ImageNet accuracy no longer transfers to new test sets—the divergence is invisible from inside the engine. A camera can be checked against an external standard; an engine reshapes the standard against which it would be checked.

Our analysis of 22,868 ICLR submissions across 2023–2025 adds a further recursive dynamic: the peer review system itself exhibits performative properties. The aggregate compliance–acceptance association (+13.0 pp in 2023, +10.7 pp in 2024, +15.8 pp in 2025) is a Simpson's paradox. Score-stratified analysis shows the within-stratum effects are small and migrate between strata across years; matched-pairs analysis yields a non-significant +2.1 pp; logistic regression shows near-zero compliance coefficient. The compliance incentive is growing, but it operates through compositional effects rather than genuine evidential improvement—precisely the structural condition under which selective reporting is most consequential.

The implication is not that ML should abandon benchmarks, ablations, or seed averages. These are useful instruments. The problem is that their social function has outgrown their statistical interpretation. A sociology of ML evaluation should ask not only whether a result is correct, but how the conditions for recognising it as correct were assembled. Benchmarks are engines because they answer these questions in practice—they do not simply record what the field knows. They help produce what the field treats as knowledge.

Broader Impact

This paper critiques evaluation practices in machine learning research. The findings could potentially be misused to delegitimise specific research groups or papers; we wish to be explicit that the argument is structural, not attributional—the evaluation pathologies we document are the predictable output of incentive systems, not evidence of individual dishonesty. The primary intended impact is to provide a sociological vocabulary for evaluation reform discussions that currently lack theoretical grounding. The structural interventions proposed (comparison logs, proof-culture-aware reviewing, venue-level requirements) are intended to be constructive and applicable across the community, including to the authors' own work.

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A Comparison Count Audit of Landmark ML Papers

Table **??** reports a manual audit of implicit comparisons in 12 landmark ML papers. For each results table, we count pairwise row comparisons within each metric column—the number of model-vs-model contrasts a reader could extract. The median paper contains 414 implicit comparisons. CLIP, which includes a 65-model linear probe across 27 datasets, contains over 58,000. These counts are lower bounds: unreported hyperparameter sweeps, architecture searches, and preliminary experiments are not captured.

Of the 12 papers, 11 (92%) mention zero multiple comparison corrections. The lone exception (CLIP) mentions correction in the context of a single experiment, not as a systematic practice across its 58,632 comparisons.

Table 5: Comparison count audit of 12 landmark ML papers. m = implicit comparisons from published table audit (pairwise row comparisons within each metric column); $\alpha^*_{\text{Bonf.}}$ = Bonferroni-corrected threshold at $\alpha = 0.05$; "Corr." = paper mentions any multiple comparison correction.

Paper	m	$\alpha^*_{\text{Bonf.}}$	Corr.
CLIP	58,632	8.5×10^{-7}	Yes
LoRA	3,301	1.5×10^{-5}	No
ViT	1,790	2.8×10^{-5}	No
LLaMA	1,559	3.2×10^{-5}	No
Attention Is All You Need	576	8.7×10^{-5}	No
ResNet	501	1.0×10^{-4}	No
BERT	414	1.2×10^{-4}	No
Mixtral	243	2.1×10^{-4}	No
DDPM	216	2.3×10^{-4}	No
GPT-2	166	3.0×10^{-4}	No
DPO	20	2.5×10^{-3}	No
SAE (Bricken et al.)	19	2.6×10^{-3}	No
Median	**414**		

B Feature Selection Audit of Mechanistic Interpretability Papers

In MI research using sparse autoencoders, the dominant comparison procedure is *feature selection*: training a dictionary of N features, examining them for interpretability, and reporting the $k \ll N$ that appear most interesting. Table **??** presents a manual audit of six MI papers.

Table 6: Feature selection audit of MI papers. "Dict N" = total features in SAE dictionary; "Shown" = features deeply examined; "Feature m" = implicit comparisons from feature selection; "Table m" = comparisons from standard result tables; α^* = Bonferroni threshold on total m. Selection ratios range from ∼200:1 to ∼3.4M:1.

Paper	Dict N	Shown	Feature m	Table m	Total m	α^*
Scaling Monosemanticity	34M	∼10	34M	0	34M	1.5×10^{-9}
Bills et al. 2023	307K	∼10	307K	4	307K	1.6×10^{-7}
Sparse Feature Circuits	33K	86	33K	84	33K	1.5×10^{-6}
SAEs Find Interp. Features	1K	∼5	1K	2	1K	4.9×10^{-5}
Towards Monosemanticity	4K	∼4	4K	19	4K	1.2×10^{-5}
ACDC	—	—	0	219	219	2.3×10^{-4}

We do not argue that SAE features are uninterpretable. We argue that the evidence base for interpretability claims is weaker than it appears because the denominator is never reported—and that the fix (reporting the interpretability *rate*, not just the highlighted examples) is technically straightforward.

C Empirical Data Sources

The quantitative findings in this paper draw on four data sources: (1) the complete ICLR 2025 (n = 11,672), ICLR 2024 (n = 7,404), and ICLR 2023 (n = 3,792) submission corpora, including reviewer scores, retrieved via the OpenReview API; (2) manual comparison-count audits of 12 landmark ML papers and 6 mechanistic interpretability papers; (3) citation-based trend analysis from OpenAlex (**?**), tracking benchmark lifecycle curves and statistical correction adoption rates via structured citation metadata; and (4) a 688-paper cross-community citation corpus (1,207 edges) assembled via Semantic Scholar, with permutation tests, temporal trend analysis, author overlap, co-citation, and reference overlap analyses. All scraping scripts, raw data, and analysis notebooks are available at `https://github.com/anonymous/benchmarks-engines` (to be de-anonymised on acceptance).

D ICLR Review Extraction Methodology

Experiment demands were extracted from reviewer text using keyword patterns in the following categories:

- **Ablation requests**: "ablation," "ablate," "should test without," "remove [component] and compare"
- **Seed/variance demands**: "run with multiple seeds," "report standard deviation," "error bars," "confidence interval"
- **Baseline additions**: "compare with," "add [method] as baseline," "missing baseline," "should also compare"
- **Dataset expansion**: "test on [other dataset]," "additional dataset," "generalize to"

Compliance was detected from author rebuttals containing phrases such as "we added," "we have updated," "new experiments," "additional results," or "revised manuscript." This keyword-based approach has known limitations: it may miss implicit compliance (e.g., silently adding a table) or misclassify discussion of future work as compliance. Manual validation on a subsample of 50 papers yielded 84% agreement with keyword extraction. The disagreements were approximately evenly split between false positives (demand-like language in non-demand context, e.g., "we compare with" in the methods section) and false negatives (compliance not signalled in the rebuttal text). The even split suggests the extraction error does not systematically inflate or deflate the compliance count.